

PAW

LoRa-Based Pet Tracking Collar

Firmware Development & Technical Deliverables

Low-Power LoRa P2P Pet Tracking System

Version 1.0

September 2026

Target Hardware: RAKwireless RAK3172 / STM32WLE5

GPS: u-blox MAX-M10S | Accelerometer: LIS2DW12 | Radio: LoRa P2P, 868 MHz

1. Project Overview

PAW is a battery-powered pet tracking collar built around LoRa Point-to-Point (P2P) radio communication, with no cellular modem or SIM connectivity. The collar reports the animal's location by transmitting compact binary packets over a private 868 MHz LoRa link, either in response to detected motion or on a fixed periodic schedule. This document describes the firmware architecture, operating sequence, packet format, radio configuration, and power design of the collar, and lists the technical deliverables produced for this development phase.

2. System Architecture

The collar's hardware is organized around a single low-power microcontroller that arbitrates between three peripherals: an accelerometer for motion-based wake-up, a GNSS receiver for position, and a LoRa radio for transmission. Battery voltage is sampled internally by the MCU.

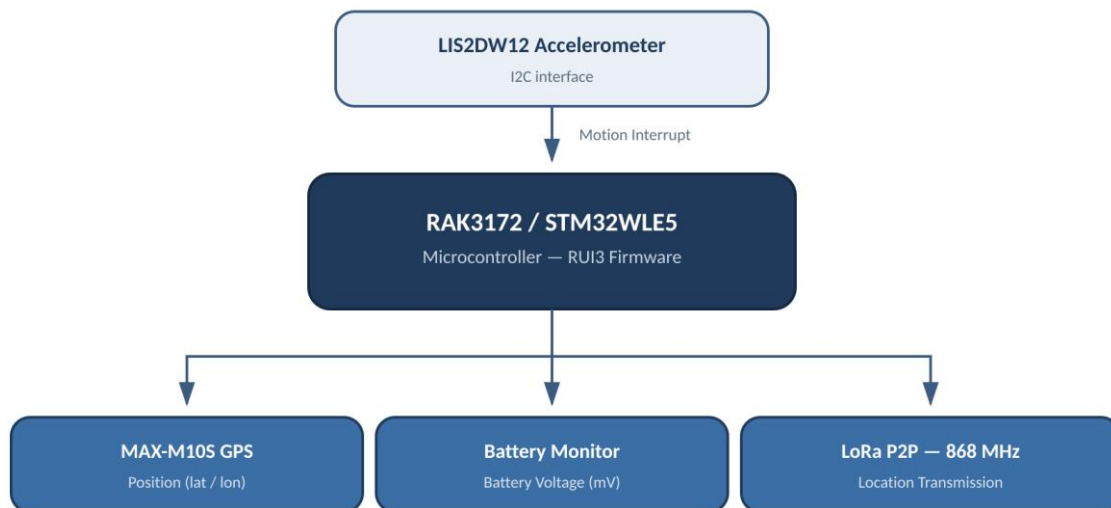


Fig. 1 — System architecture: MCU coordinates GPS, battery monitoring, and LoRa P2P transmission.

3. Firmware Operating Sequence

On each wake event, the firmware executes the following sequence before returning to low-power sleep:

- Low-power sleep — MCU remains in its lowest power state between events.
- Motion interrupt or periodic wake-up — the accelerometer's motion interrupt, or the 30-minute periodic timer, wakes the MCU.
- GPS acquisition — the GNSS receiver is powered/queried to obtain a position fix, bounded by a fix timeout.
- GPS validation — a fix is accepted only if the reported fix quality is valid; failed fixes are logged and the payload's GPS-valid flag is cleared.
- Battery measurement — the MCU samples its battery voltage for inclusion in the payload.
- Binary payload construction — device ID, position, timestamp, battery voltage, and a sequence number are packed into a fixed-length binary payload.
- LoRa P2P transmission — the payload is transmitted over the configured 868 MHz P2P link.

- Return to low-power sleep — radio and non-essential peripherals are left idle and the MCU resumes deep sleep until the next event.

4. Professional State Machine Diagram

The diagram below shows the firmware's state machine. Two independent wake-up sources — the accelerometer motion interrupt and the 30-minute periodic fix timer — both enter the same WAKE / EVENT DETECTION state, after which the sequence is identical regardless of which source triggered it.

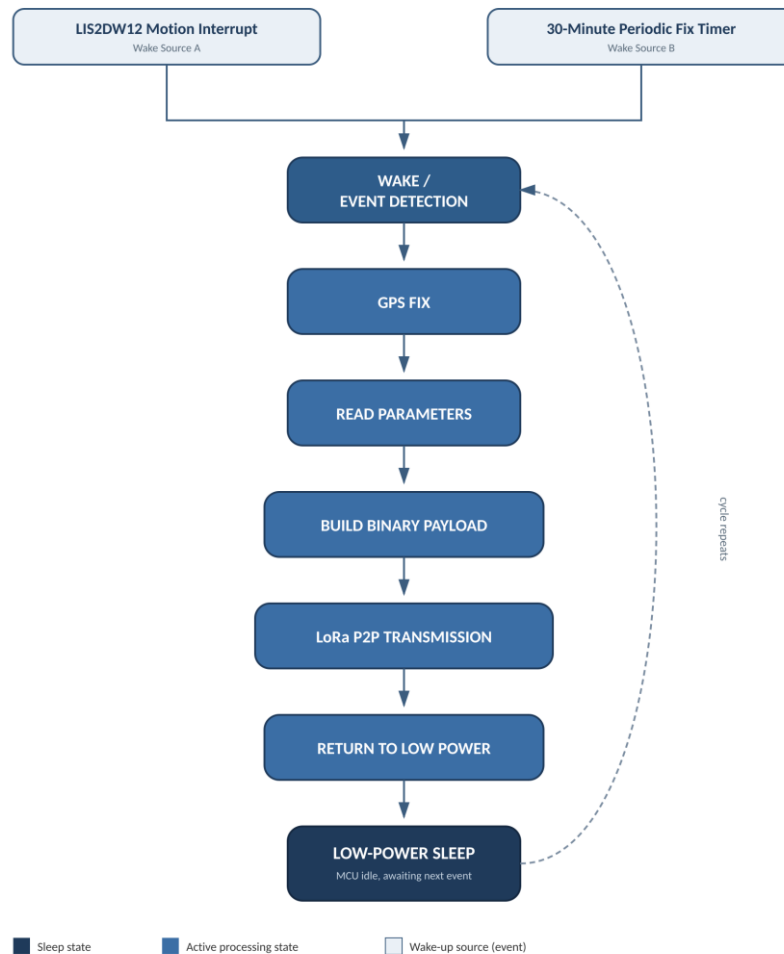


Fig. 2 — Collar firmware state machine, showing both wake-up paths converging on a common processing sequence.

5. Binary Packet Format

Each location report is encoded as an 18-byte binary payload using little-endian byte order.

Byte	Field	Type	Description
0	Device ID	uint8	Unique collar identifier
1	Protocol Version	uint8	Packet format version
2	Status Flags	uint8	Bit 0: GPS valid Bit 1: motion-triggered wake
3–6	Latitude	int32	Degrees × 1,000,000

Byte	Field	Type	Description
7–10	Longitude	int32	Degrees × 1,000,000
11–14	Timestamp	uint32	Seconds since boot
15–16	Battery	uint16	Battery voltage, millivolts
17	Sequence	uint8	Packet sequence number (wraps at 256)

Total payload size: 18 bytes. Encoding: little-endian.

6. Firmware Implementation

The firmware is organized around hardware-abstraction functions for each peripheral subsystem, so the control logic (wake handling, fix acquisition, payload construction, transmission) is decoupled from the specific driver calls beneath it:

- Accelerometer — LIS2DW12 initialization, interrupt configuration, and interrupt-source clearing.
- GPS — NMEA GGA sentence acquisition and parsing over UART.
- Battery measurement — MCU battery voltage sampling.
- LoRa P2P — radio configuration and packet transmission.
- Low-power management — sleep mode entry and low-power level configuration.

Implementation note: GPS acquisition supports a mock mode for development and bench testing when physical GNSS hardware is unavailable, generating a simulated fix in place of live NMEA data. The LoRa radio path is unaffected by this switch and remains configured for real P2P transmission at all times. Real NMEA GGA parsing is implemented and used automatically once GPS hardware is attached and mock mode is disabled.

7. LoRa Configuration

Parameter	Value
Frequency	868 MHz
Mode	LoRa P2P
Spreading Factor	SF9
Coding Rate	4/5
Preamble	8 symbols
TX Power	14 dBm

8. Power Optimization Strategy

The firmware's power strategy is event-driven rather than polling-based: the MCU remains in a low-power mode by default and is only woken by the accelerometer's motion interrupt or by the periodic fix timer, avoiding any continuous polling loop that would keep the processor active. The GPS receiver is activated only

when a location fix is actually required, rather than being left powered continuously, and the LoRa radio is enabled only for the duration of a transmission before returning to an idle state. A motion transmission cooldown prevents repeated triggering from continuous movement, limiting how often a motion event can initiate a full acquire-and-transmit cycle. Independently of motion activity, a maximum 30-minute periodic update guarantees the collar is not silent for extended periods, while still keeping the MCU in low-power sleep between events. After each transmission the MCU disables non-essential peripherals and returns to low-power mode.

Engineering estimate: Any battery-life figure for this design depends on the specific battery capacity used and on measured average current draw across sleep and active states, neither of which has been characterized on physical hardware at this stage. A battery-life number is not stated here; it should be calculated once a battery capacity is selected and current draw is measured or modeled against the actual fix rate in deployment, and clearly labeled as an estimate when reported.

9. Firmware State Description

State	Function
LOW-POWER SLEEP	MCU remains in low-power mode, awaiting a wake-up event.
WAKE	Detect and identify the wake-up source (motion interrupt or periodic timer).
GPS FIX	Acquire and validate a GNSS position fix within the fix timeout.
READ PARAMETERS	Read battery voltage and event context (motion vs. periodic).
BUILD PAYLOAD	Encode position, timestamp, battery, and sequence data into the binary payload.
LoRa TX	Transmit the binary packet over the configured LoRa P2P link.
RETURN TO SLEEP	Disable unnecessary peripherals and resume low-power mode.

10. Deliverables Summary

The following technical deliverables accompany this document:

- Firmware Source Code — RUI3-based collar firmware with hardware-abstraction layer and mockable GPS.
- State Machine Diagram — sleep / wake / fix / transmit / sleep cycle with both wake-up paths shown.
- Binary Packet Specification — 18-byte, little-endian payload format.
- LoRa Configuration — 868 MHz P2P radio parameters (SF9, CR 4/5, 8-symbol preamble, 14 dBm).
- Power Optimization Strategy — event-driven design rationale, with battery life left as an open engineering estimate pending measured current data.